Determinants of Basketball Player Performance: A Physical-Attributes Approach

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1 Introduction

The goal of this project is to analyze basketball player performance and identify the factors that most influence it. Specifically, we focus on NBA players, using detailed player statistics and physical attributes to explore performance trends and build predictive models. This project anchors itself in the high-stakes reality of modern sports management. In a multi-billion dollar industry where recruitment errors are costly, General Managers and scouts increasingly rely on data to move beyond the traditional "eye test." Our analysis aims to provide actionable insights for optimizing roster construction and contract negotiations.

Our research questions are centered on identifying which variables are most predictive of performance. In particular, we aim to understand how players and their performances evolve over time, how physical characteristics influence effectiveness, and which metrics best predict scoring and overall contribution to the game. In addition, we develop a predictive model designed to identify the combination of player features that leads to optimal efficiency.

This project builds on our original proposal by incorporating a broader dataset covering 49 leagues and over 11,000 players from 1999 to 2020. The dataset includes detailed performance statistics, contextual information (league, season, stage), biometric data (height, weight, birth year), and background information (draft, high school, nationality). Our initial focus was on exploratory data analysis (EDA) to understand the structure and key features of the data. After reviewing the dataset, we refined our approach to include both EDA and predictive modeling, allowing us to quantify relationships between variables and estimate future player performance.

2 Data

2.1 Data sources

Our project uses two primary datasets found on kaggle.com:

players_stats_by_season_full_details.xlsx (Baruch 2023), which covers professional basketball seasons from 1999 to 2020 across 49 leagues, representing around 11,000 players. It contains detailed information such as birth dates, height, weight, nationality, and high school background, along with comprehensive season statistics (points, rebounds, assists, shooting percentages, minutes, games played, and more). This dataset offers much more than raw numbers: it provides a broad view of global player performance over more than two decades, making it a strong foundation for comparative and longitudinal analysis.

NBA_Anthropometric.csv (Dobrucki 2023), which contains anthropometric measurements from NBA Draft Combine events held between 2000 and 2023. The NBA Draft Combine is a yearly showcase where invited prospects undergo physical testing, medical exams, athletic drills, shooting evaluations, and five-on-five scrimmages in front of NBA coaches, general managers, and scouts. But this file includes only a limited set of physical measurements (height, weight, wingspan, arm length, etc.) converted from imperial to metric units, representing just a small part of all data collected at the Combine. The data was originally obtained through the NBA Stats API and converted from imperial to metric units.

Both datasets were loaded into Python and inspected to understand their structure.

```
Data loaded successfully.
```

2.2 Data cleaning and pre-processing

The two datasets did not perfectly match (different naming formats, missing players, duplicates). As a first step, we identified and kept only the players present in both datasets.

```
Merged rows: 6366
```

We also ensured consistent player identification across both sources by normalizing the player name fields and comparing unique values, which prevented duplicated entries and guaranteed that each player-season appears only once in the final dataset. In addition, several columns (e.g., height_stats, birth_month, birth_year, player_name, height_cm) were removed because they were either duplicated across datasets, irrelevant for performance modeling, inconsistent, or missing for many players.

We also addressed inconsistencies in date formats within the birth_date column to enable precise age calculations. Since many players appear in the dataset across multiple seasons, relying on a static age would be inaccurate. We therefore calculated a dynamic age for each season entry by computing the time difference between the player's normalized birth date and the start of that specific season (fixed at October 1st). This step was crucial for correctly analyzing the "aging curve" and performance evolution throughout a player's career.

To make players comparable despite differences in playing time, we normalized all box-score statistics on a per-minute basis. This step was necessary because raw totals (e.g., points, rebounds, assists) are heavily influenced by the number of minutes a player spends on the court. By creating variables such as `PTS_per_min`, `REB_per_min`, `AST_per_min`, and all other equivalent per-minute metrics, we ensured that the analysis reflected true performance intensity rather than total playing time.

To evaluate overall performance, we adopted the standard NBA Efficiency (EFF)(Captain Calculator 2025) metric but normalized it to a per-minute basis. While standard per-game EFF scores typically range from 10 to 30, our resulting `EFF_per_min` metric scales down to a range of 0 to 1.5 . The metric is calculated by summing a player's positive contributions (points, rebounds, assists, steals, blocks) and subtracting the negative ones (missed shots, missed free throws, and turnovers), all divided by minutes played. To interpret this scale: a value of 0.5

EFF/min represents a solid rotation player, while anything >1.0 EFF/min indicates elite, MVP-level productivity.

```
Colonnes créées : ['FGM_per_min', 'FGA_per_min', '3PM_per_min', '3PA_per_min', 'FTM_per_min'
```

The `position` column often contains dual entries (e.g., "SF/SG"). We chose to keep only the **first position** listed. In basketball reporting, the first position typically indicates the player's primary defensive assignment and offensive role, while the second indicates secondary versatility. Simplifying to the primary role allows for clearer categorical analysis.

For the remainder of this project, the analysis focuses exclusively on the NBA. This choice is motivated by the league's consistent level of competition, standardized rules, and high-quality data availability. Restricting the scope to the NBA allows for more reliable comparisons between players and seasons, while reducing variability introduced by differences across leagues.

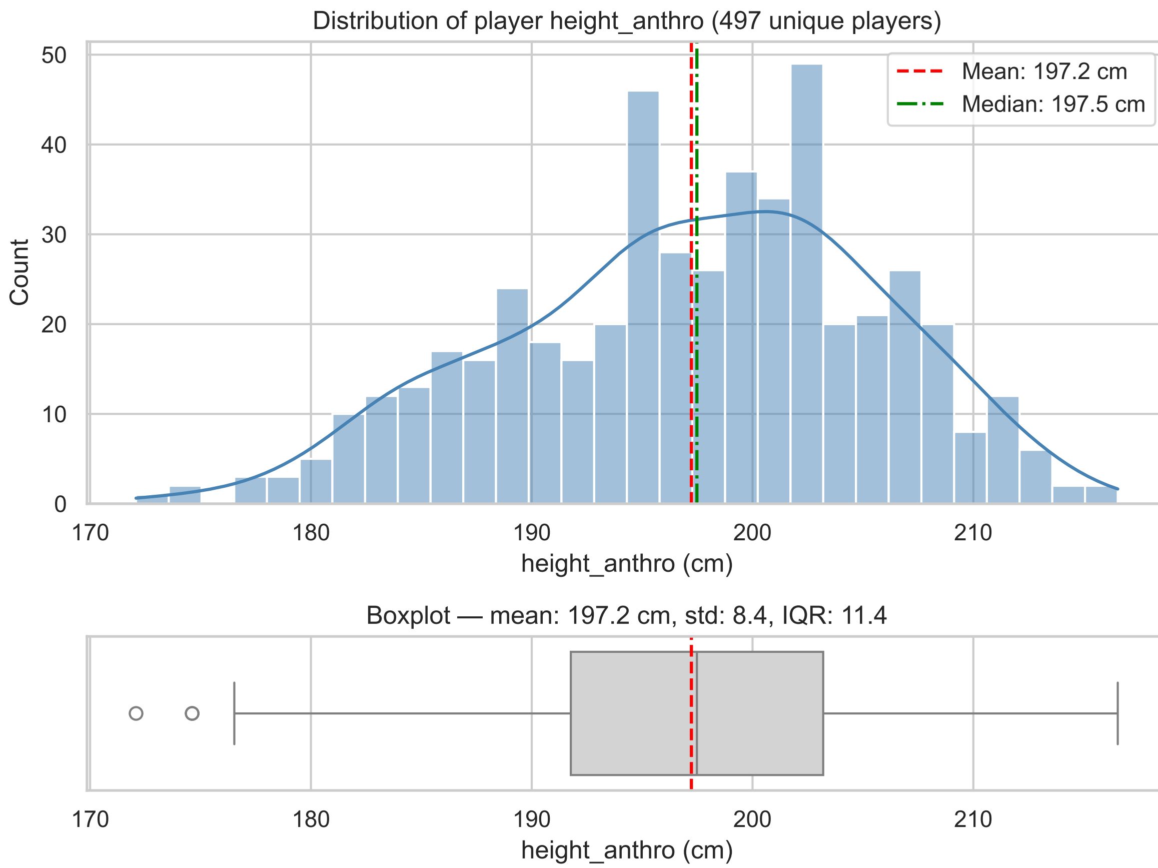
2.3 Main challenges

The main challenges were merging the two datasets and eliminating duplicate columns. We considered integrating the teams each player played for, but creating dummy variables for all teams made the system cumbersome and difficult to use. Beyond the technical complexity, integrating the teams is particularly irrelevant in our case. We don't have a complete sample of players for each team, and we don't know precisely when players play together during the season.

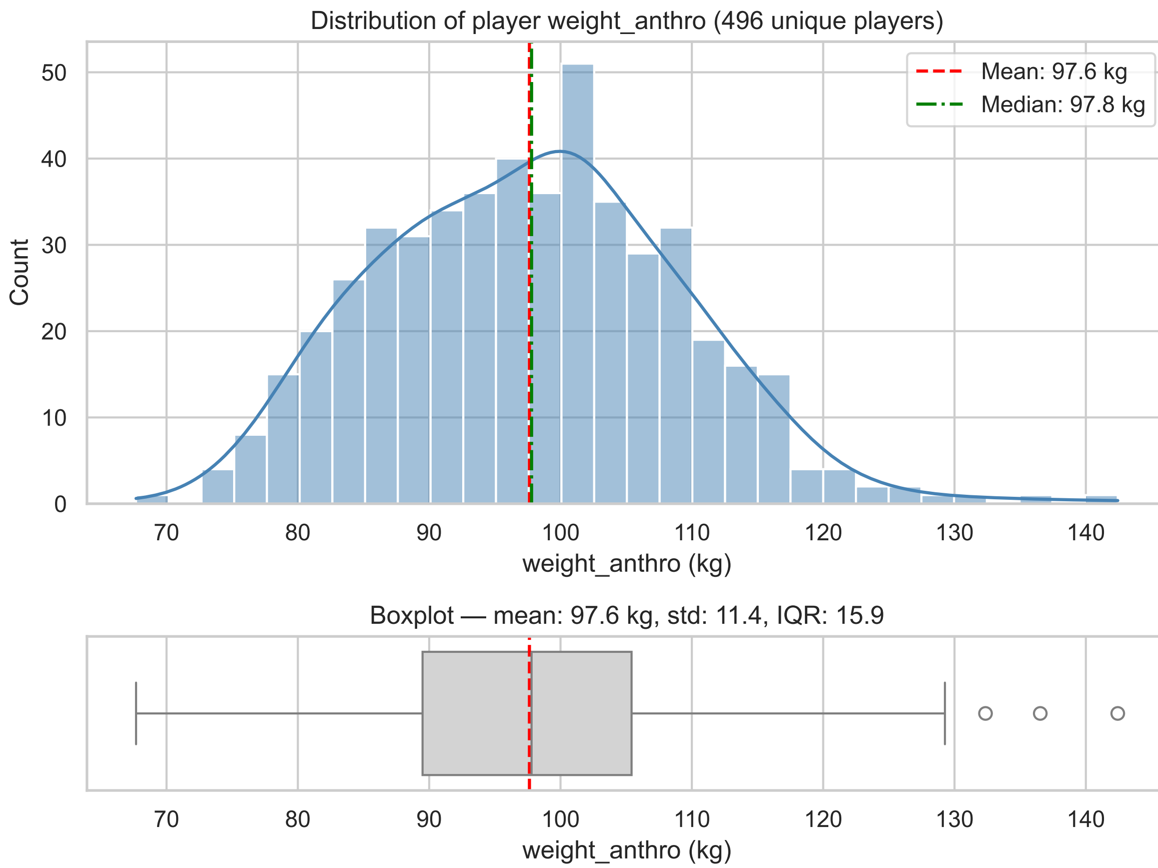
3 Exploratory Data Analysis (EDA)

3.1 Summary statistics

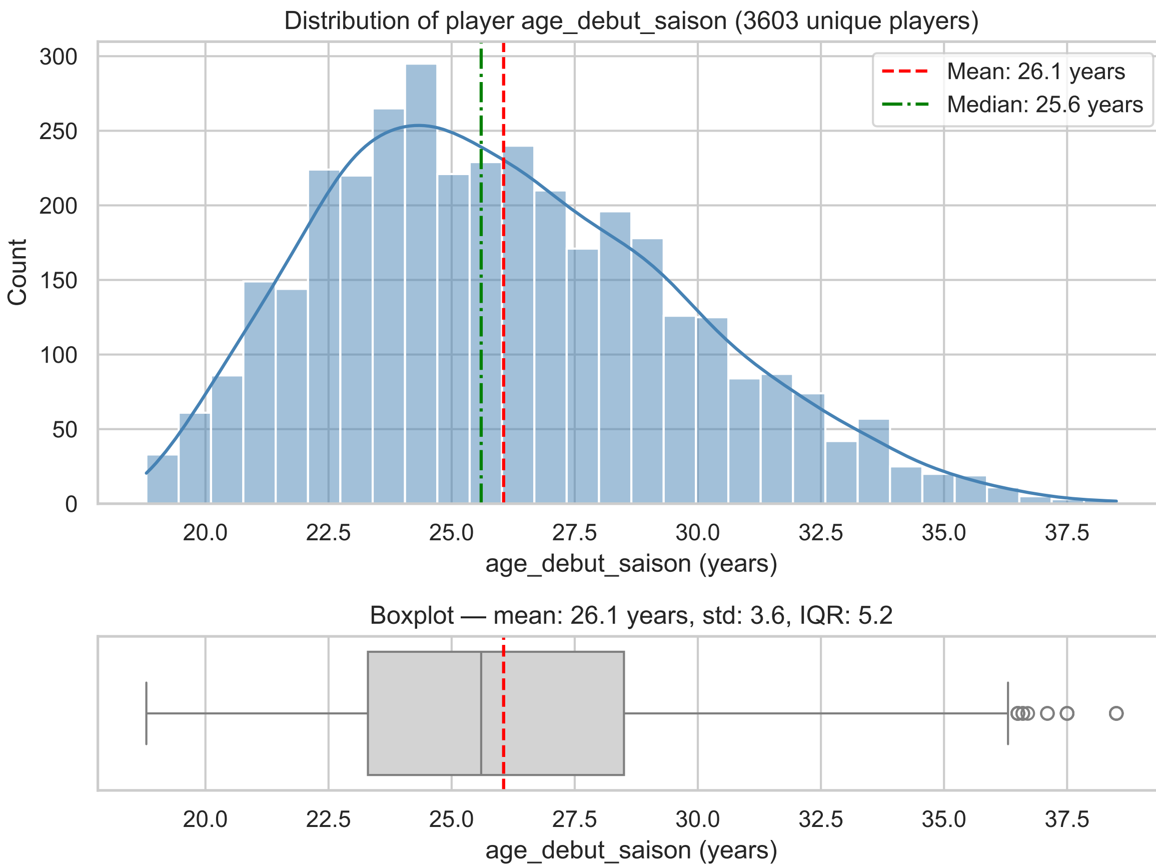
We began our analysis by examining player physical characteristics, focusing first on height, weight, age at the start of the season, and EFF per minute. Using histograms and boxplots (considering each player only once), we explored the distribution of these stats among all players in the dataset. Summary statistics indicated the average, median, standard deviation, and interquartile range of for each player, while extreme values were flagged as potential outliers. This step allowed us to identify unusual or inconsistent entries and better understand the baseline characteristics of our population.



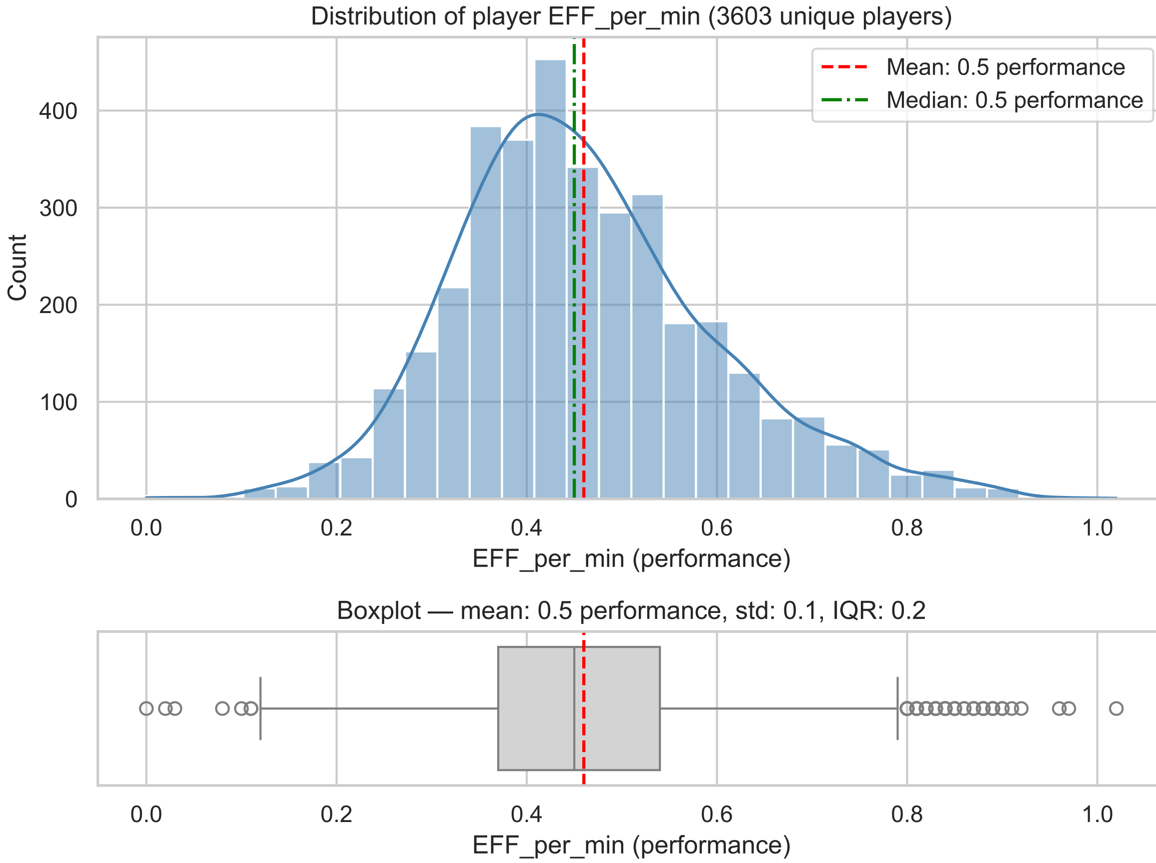
HEIGHT_ANTHRO - Count: 497, Mean: 197.2 cm, Median: 197.5 cm, Std: 8.4, IQR: 11.4




```
WEIGHT_ANTHRO - Count: 496, Mean: 97.6 kg, Median: 97.8 kg, Std: 11.4, IQR: 15.9
```



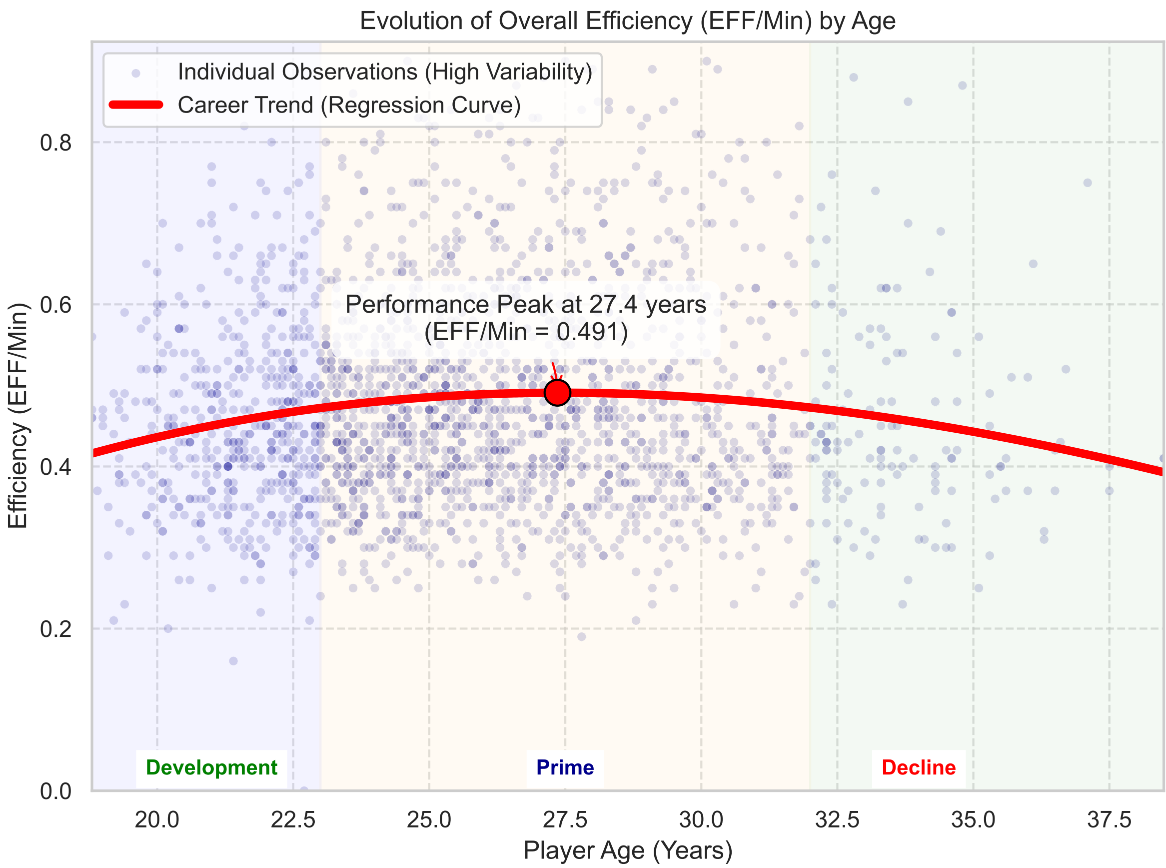

```
AGE_DEBUT_SAISON - Count: 3603, Mean: 26.1 years, Median: 25.6 years, Std: 3.6, IQR: 5.2
```




```
EFF_PER_MIN - Count: 3603, Mean: 0.5 performance, Median: 0.5 performance, Std: 0.1, IQR: 0.
```

Here we visualize the target variable `EFF_per_min`. As explained in the preprocessing, the values are small. The distribution is somewhat normal but with a right skew. Which indicates that "superstar" productivity is rare. Most players cluster around the 0.4-0.6 range.

We also analyzed how player performance changes with age by looking at per-minute efficiency for players who played multiple seasons. Using a polynomial trend line, we were able to see the general evolution of performance across a career and identify the age at which players typically reach their peak. This analysis is important because it shows not just individual variation but the overall pattern of development and decline, helping us understand typical career trajectories in the NBA.



3.2 Key findings

First, the distributions of height, weight, and age confirmed the expected structure of an NBA population: players are generally very tall, with most heights clustered around similar ranges, and age is centered around the mid-20s. The boxplots also helped us spot a few outliers in height and weight, which could indicate unusual player profiles or data inconsistencies.

When looking at efficiency across ages, the curve we plotted showed a clear performance peak at around 27.4 years old. This aligns with the idea that players combine both physical prime and solid experience at that age. Before that, efficiency tends to increase gradually as players develop, and after the peak, we observed a slow decline that matches the natural physical aging curve. Even though this is just a broad trend, it gives a first impression of how career progression typically looks in the NBA.

4 Analysis

For the analysis, we focused on a few statistical models to better understand which factors actually drive player performance. We mainly used simple and multiple linear regressions to see how variables like height, weight, wingspan, age, and position relate to `EFF_per_min` when looked at together. These models help us go beyond basic trends from the EDA and test whether the patterns we saw still hold once variables are combined.

In a second step, we introduced polynomial regressions to capture potential non-linear effects between physical attributes and performance. Finally, we built a predictive model that combines anthropometric variables and player position to estimate efficiency per minute and explore theoretical optimal player profiles under realistic physical constraints.

4.1 Simple linear regressions

After that, we explored the relationship between player attributes and performance using simple linear regression models. Specifically, we regressed `EFF_per_min` on individual physical or demographic variables, such as height. This approach allowed us to quantify how each attribute influences overall efficiency per minute and to assess the statistical significance of the effect. For example, regressing efficiency on height helped determine whether taller players tend to be more effective on a per-minute basis. By running these regressions for several key variables, we identified which attributes are significant predictors of performance and which have little to no measurable effect.

```
Linear Regression
Dependent Variable : EFF_per_min
Explicatives Variables : height_anthro
```

	Coefficient	Std. Error	p-value	
const	-0.314	0.053	0.0	***
height_anthro	0.004	0.0	0.0	***
R-squared	0.056			
Observations	3588			

The regression shows a positive and statistically significant effect of height on efficiency per minute, with an estimated coefficient of 0.004 ($p < 0.001$), but the model has low explanatory power, with an R-squared of 0.056 (N = 3,588).

```
Linear Regression
Dependent Variable : EFF_per_min
Explicatives Variables : wingspan
```

	Coefficient	Std. Error	p-value	
const	-0.396	0.046	0.0	***
wingspan	0.004	0.0	0.0	***
R-squared	0.090			
Observations	3593			

The regression shows a positive and statistically significant effect of wingspan on efficiency per minute, with an estimated coefficient of 0.004 ($p < 0.001$), but the model has limited explanatory power, with an R-squared of 0.090 (N = 3,593).

```
Linear Regression
Dependent Variable : EFF_per_min
Explicatives Variables : weight_anthro
```

	Coefficient	Std. Error	p-value	
const	0.16	0.019	0.0	***
weight_anthro	0.003	0.0	0.0	***
R-squared	0.064			
Observations	3586			

The regression shows a positive and statistically significant effect of weight on efficiency per minute, with an estimated coefficient of 0.003 ($p < 0.001$), but the model has low explanatory power, with an R-squared of 0.064 (N = 3,586).

```
Linear Regression
Dependent Variable : EFF_per_min
Explicatives Variables : age_debut_saison
```

	Coefficient	Std. Error	p-value	
const	0.457	0.017	0.0	***

	Coefficient	Std. Error	p-value
age_debut_saison	0.0	0.001	0.859
R-squared	0.000		
Observations	3603		

The regression shows that age at season debut has no statistically significant effect on the dependent variable (p = 0.859), and the model has virtually no explanatory power, with an R-squared close to zero, despite a large sample size (N = 3,603).

4.2 Multiple linear regressions

After analyzing individual variables with simple regressions, we then performed multiple linear regressions to see how these factors together influence scoring efficiency. For example, we modeled `EFF_per_min` as a function of height, weight, age, and the interaction between height and age. We did that including the position of the player as well, which influences quite a lot how much a player scores, since different positions have different typical roles and responsibilities on the court.

```
 Multiple Linear Regression Results (EFF_per_min vs Player Profile)
Model: EFF_per_min = B0 + B1*Height + B2*Weight + B3*Age + B4*(Height * Age)
```

	Coefficient	Std. Error	p-value	
Intercept	-0.657	0.471	0.163	
height_anthro	0.005	0.002	0.06	*
weight_anthro	0.002	0.0	0.0	***
age_debut_saison	0.02	0.018	0.263	
height_anthro:age_debut_saison	-0.0	0.0	0.309	
R-squared	0.088			
Observations	2251			

The first model, which excludes player position, shows that both height and weight are significant predictors, with positive coefficients suggesting that taller and heavier players tend to have higher efficiency per minute (`EFF_per_min`). However, Age and its interaction with height are found to be statistically insignificant. This model only accounts for a small amount of the variance in efficiency, as indicated by a low **R-squared** of **0.088**.

```
 Multiple Linear Regression Results (EFF_per_min vs Player Profile + Position)
Model: EFF_per_min = f(Height, Weight, Age, Height * Age, Position)
```

	Coefficient	Std. Error	p-value	
Intercept	-0.461	0.469	0.327	
C(Simplified_Position)[T.PF]	-0.022	0.011	0.048	**
C(Simplified_Position)[T.PG]	-0.011	0.021	0.609	
C(Simplified_Position)[T.SF]	-0.084	0.013	0.0	***
C(Simplified_Position)[T.SG]	-0.084	0.016	0.0	***
height_anthro	0.004	0.002	0.091	*
weight_anthro	0.001	0.0	0.0	***
age_debut_saison	0.013	0.017	0.468	
height_anthro:age_debut_saison	-0.0	0.0	0.548	
R-squared	0.160			
Observations	2251			

The second, more powerful model (with **R-squared=0.160**) confirms that Position is a major determinant of efficiency. The significant and negative coefficients for PF, SF, and SG (relative to the unlisted reference position C) indicate that these positions typically have lower `EFF_per_min`.
Although the exploratory analysis suggests a clear age-related performance pattern, age is not statistically significant in the multiple regression models once physical attributes and position are controlled for. This indicates that age captures broad career trends rather than a direct effect on efficiency per minute.

4.3 Regular season vs playoffs

We finally compared performance in the regular season with that in the playoffs. This comparison is particularly important because playoffs games carry higher stakes: players are expected to perform at their best, yet they also face greater pressure.

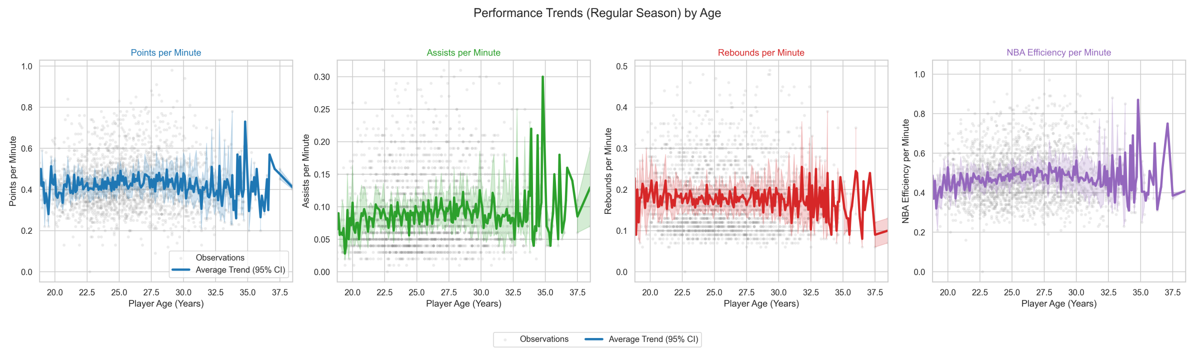


Figure 1: Regular Season Performance

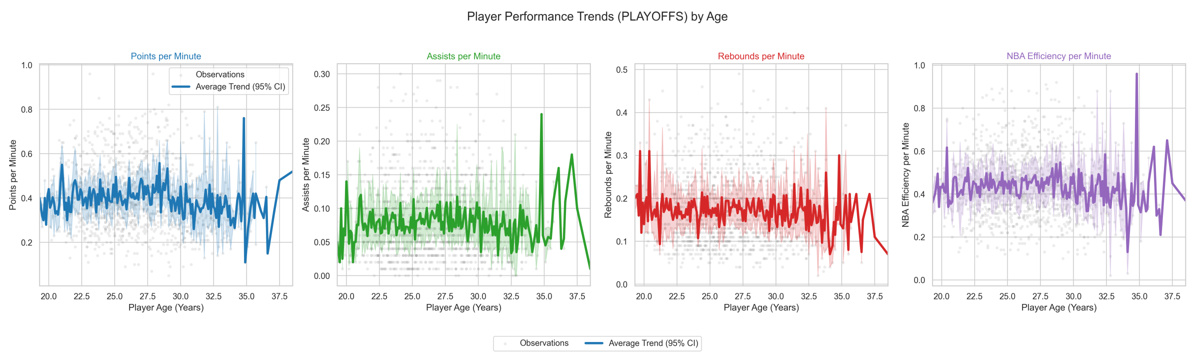


Figure 2: Playoffs Performance

Regular season vs. playoffs performance

Overall, players exhibit reduced offensive efficiency in the playoffs. Declines in points, assists, and overall efficiency suggest that the heightened defensive intensity and pressure of postseason competition limit scoring opportunities and offensive flow. Defensive metrics, such as steals and rebounds, also decrease slightly, reflecting the more physical and strategic nature of playoffs games. Metrics related to rare events, like missed shots or blocks, remain largely unchanged, indicating that shooting accuracy and occasional defensive actions are less affected by the stage.

	Metric	Regular Season Mean	Playoffs Mean	Mean Diff (PO - RS)	p-value
1	Assists per Min (Offense)	0.0886	0.0802	-0.0084	0.0000
3	Efficiency per Min (Overall)	0.4771	0.4454	-0.0317	0.0000
6	Turnovers per Min (Negative)	0.0556	0.0525	-0.0031	0.0005
0	Points per Min (Offense)	0.4252	0.4079	-0.0173	0.0006
4	Steals per Min (Defense)	0.0317	0.0299	-0.0018	0.0008

	Metric	Regular Season Mean	Playoffs Mean	Mean Diff (PO - RS)	p-value
2	Rebounds per Min (Offense)	0.1773	0.1704	-0.0070	0.0177
8	Missed FTs per Min (Negative)	0.0238	0.0250	0.0012	0.1325
5	Blocks per Min (Defense)	0.0201	0.0195	-0.0006	0.4234
7	Missed FGs per Min (Negative)	0.1864	0.1850	-0.0014	0.5542

Elite players and playoffs performance

Elite players defined as the top 10% in regular-season efficiency per minute tend to be able to maintain, their efficiency during the playoffs. This suggests that top performers are better able to sustain high-level production under increased competitive pressure.

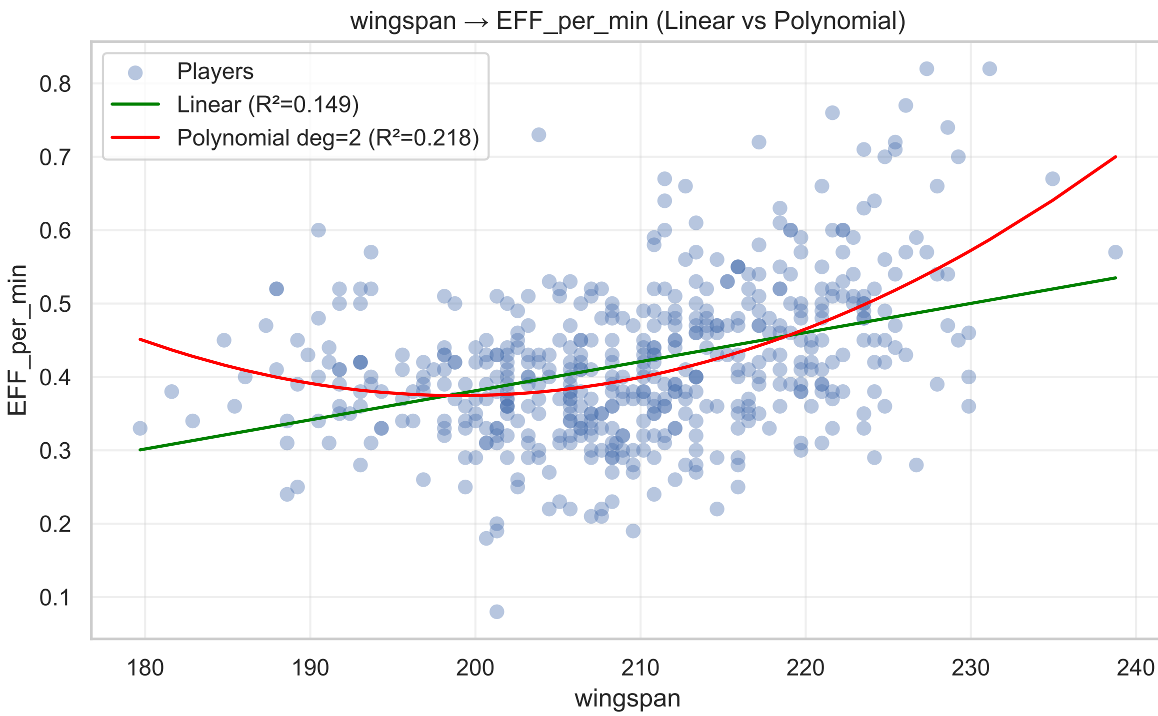
	Metric	Value
0	Elite threshold (90th percentile, Reg. Season)	0.6094
1	Number of elite players	49.0000
2	Avg EFF/min (Reg. Season, Elite)	0.6843
3	Avg EFF/min (Playoffs, Elite)	0.6081
4	Avg differential (Playoffs − Reg. Season)	-0.0762

Stage	Player	Avg_EFF_Reg_Season_Full	Avg_EFF_Playoffs_Full	EFF_Differential
221	Jarrett Allen	0.6600	0.7400	0.0800
181	Greg Monroe	0.6286	0.6450	0.0164
279	Kawhi Leonard	0.6662	0.6725	0.0062
306	LeBron James	0.7582	0.7493	-0.0089
200	JaVale McGee	0.6812	0.6680	-0.0133

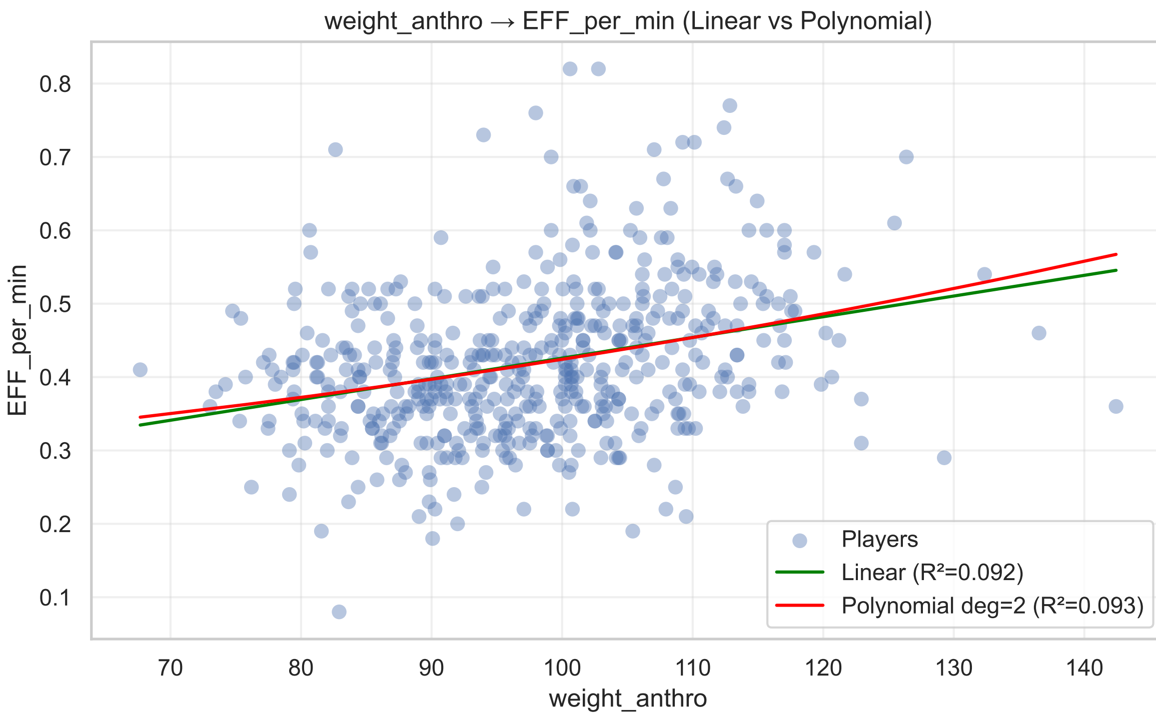
Overview of the 5 players with the smallest EFF differential.

4.4 Polynomial regressions

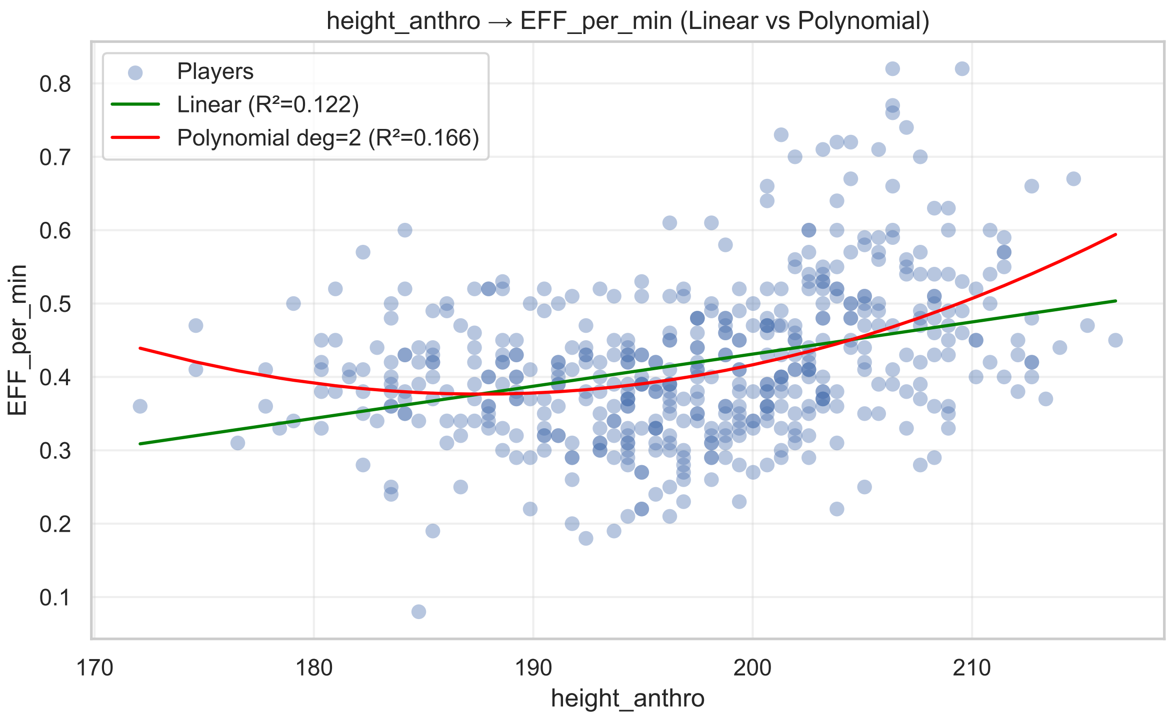
We chose to compare linear regression with polynomial regression to determine if we could develop a more accurate prediction model.



wingspan → EFF_per_min (Linear vs Polynomial)
Players
Linear (R²=0.149)
Polynomial deg=2 (R²=0.218)
EFF_per_min
wingspan



weight_anthro → EFF_per_min (Linear vs Polynomial)
Players
Linear (R²=0.092)
Polynomial deg=2 (R²=0.093)
EFF_per_min
weight_anthro



We noticed an interesting difference in the **R-squared** predictions for both wingspan and height. We can now proceed to perform a multifactor polynomial regression.

```
Number of observations: 3575
```

Table 10: Polynomial Regression Results (Intercept = EFF of an average player)

	Coefficient	Std. Error	t-statistic	P-value	CI 2.5%	CI
Intercept	0.4200	0.0152	27.6134	0.0000	0.3902	0.4
poly_features__height_anthro	0.0021	0.0054	0.3899	0.6967	-0.0084	0.0
poly_features__wingspan	0.0412	0.0043	9.6292	0.0000	0.0328	0.0
poly_features__weight_anthro	0.0226	0.0041	5.4879	0.0000	0.0145	0.0
poly_features__height_anthro^2	0.0025	0.0060	0.4138	0.6791	-0.0093	0.0
poly_features__height_anthro wingspan	0.0005	0.0100	0.0509	0.9594	-0.0192	0.0
poly_features__height_anthro weight_anthro	0.0114	0.0073	1.5504	0.1211	-0.0030	0.0
poly_features__wingspan^2	0.0216	0.0062	3.4897	0.0005	0.0095	0.0
poly_features__wingspan weight_anthro	-0.0177	0.0077	-2.3017	0.0214	-0.0327	-0.
poly_features__weight_anthro^2	-0.0026	0.0037	-0.7133	0.4757	-0.0098	0.0
position_simple__position_simple_PF	0.0266	0.0113	2.3607	0.0183	0.0045	0.0
position_simple__position_simple_PG	0.0701	0.0192	3.6566	0.0003	0.0325	0.1

	Coefficient	Std. Error	t-statistic	P-value	CI 2.5%	CI
position_simple___position_simple_SF	-0.0050	0.0143	-0.3485	0.7275	-0.0329	0.0
position_simple___position_simple_SG	0.0033	0.0169	0.1924	0.8475	-0.0300	0.0

We created a centered polynomial regression model so that the intercept represents an average player with a predicted `EFF_per_min` of 0.42. This allows us to interpret the coefficients relative to a typical player. The model includes quadratic and interaction terms, which highlight subtle effects, such as the positive quadratic effect of wingspan and the interaction between wingspan and weight. These terms help identify how combinations of physical measurements influence efficiency beyond simple linear effects.

4.5 Theoretical performance model

To ensure anthropometric realism in the optimization procedure, wingspan is not treated as an independent variable. Instead, it is modeled as a deterministic function of height, using the average wingspan-to-height ratio observed in the NBA. In addition, player weight is constrained using a BMI-based plausibility range, ensuring that only realistic height–weight combinations are considered. Together, these constraints reflect biological proportionality and prevent the model from selecting physically implausible player profiles. The resulting optimal player therefore represents a realistic combination of height, weight, and position rather than a purely statistical extreme. But it should not be interpreted as a real or achievable player type, but rather as a statistical illustration of how physical attributes interact within the model under realistic constraints.

```
Extended Theoretical Optimal Player (NBA-Plausible)
height: 235.00
wingspan: 249.74
weight: 107.65
position: PG
predicted_EFF_per_min: 1.07
```

We modeled the resulting surface, which isolates the combined effect of height and weight on efficiency while preserving realistic body proportions. As we can see, there is an optimal point around **108 kg** for the tallest measurements allowed.

```
Average NBA wingspan-to-height ratio: 1.063
```

```
Unable to display output for mime type(s): text/html
```

```
Unable to display output for mime type(s): text/html
```

4.6 What we discovered

From our analysis, a few things stood out. We saw that height and experience don't play the same role for everyone. For rather short players, having more experience becomes a key factor in their performance. This suggests that being tall gives an immediate advantage, whereas shorter players depend more on skill development and game understanding.

When looking at performance over age, we identified a clear peak around 27.4 years old, which makes sense: players are still in their physical prime but already have enough experience to make better decisions. Another interesting pattern is that players who remain active at older ages usually display high EFF values. This is likely driven by a survivorship effect: only the most efficient players are able to stay in the league as they age.

We also noticed that player position explains a big part of the differences in EFF per minute. Interior players, so mainly Centers (C) and Power Forwards (PF), tend to show higher efficiency than perimeter players such as Small Forwards (SF), Shooting Guards (SG), and Point Guards (PG). This makes sense because players in inside positions are closer to the basket, take higher-percentage shots, and are more involved in rebounds and actions near the rim, which naturally boosts their efficiency.

We then compared Regular Season and Playoffs performances. We expected players to perform better in Playoffs because the games matter more, but what we observed is the opposite: almost everyone sees their EFF drop. The simplest explanation, besides higher pressure, is the increase in defensive intensity, as scoring becomes harder, mistakes are punished more, and efficiency tends to fall. Only two players in our dataset actually improved their EFF in the Playoffs; for all the others, it went down.

Lastly, we used a centered polynomial regression to better understand how players' physical characteristics and positions relate to efficiency per minute. By focusing on an average player, this approach allowed us to interpret the results more intuitively. The model shows that wingspan and weight are the most relevant physical factors, with some non-linear effects, while height alone plays a more limited role. Position also remains important, as point guards and power forwards tend to display slightly higher efficiency compared to other roles. Overall, this analysis complements our earlier findings by showing how physical attributes and positional roles help explain differences in performance beyond age and experience.

5 5. Conclusion

5.1 Summary of main findings

In this project, we analyzed NBA player performance using physical attributes, age, position, and efficiency per minute metrics. Our results show that physical characteristics such as height, weight, and especially wingspan are positively related to efficiency per minute, although their explanatory power remains limited. We also identified a clear performance peak around 27.4 years old, which reflects the balance between physical prime and accumulated experience. In addition, player position plays a major role in explaining differences in efficiency, with point guards and power forwards generally displaying higher `EFF_per_min` than other roles.

5.2 Practical implications

From a practical perspective, our findings suggest that physical attributes should be seen as advantages rather than guarantees of high performance. While being taller or heavier can improve efficiency on average, these factors alone cannot fully explain why some players perform better than others. The comparison between Regular Season and Playoffs also highlights the importance of context: most players experience a drop in efficiency during the Playoffs, while elite players tend to maintain more stable performance under pressure. This suggests that consistency and adaptability are key traits that may not be fully captured by standard physical or box-score metrics.

5.3 Limitations and future research

Despite these insights, our analysis is limited by the available data. We focused mainly on anthropometric variables and basic performance statistics, which do not capture important aspects such as athletic testing results, tactical roles, or psychological factors. Future research could integrate combine data such as speed, vertical jump, or agility, as well as tracking data related to movement and workload. Using more advanced longitudinal models could also help better understand player development over time and improve the prediction of long-term performance.

Baruch, Jacob. 2023. "Basketball Players Stats Per Season (49 Leagues)." https://www.kaggle.com/datasets/jacobbaruch/basketball-players-stats-per-season-49-leagues.

Captain Calculator. 2025. "Basketball Player Efficiency Calculator." https://captaincalculator.com/sports/basketball/efficiency/.

Dobrucki, Tymoteusz. 2023. "NBA Anthropometric Data." https://www.kaggle.com/datasets/tymoteuszdobrucki/nba-anthropometric.